

Full Control Design via Carleman Linearization and LQR for a Segway-Like System

1 Introduction

In this document, we describe the complete control design for a Segway-like system (or wheeled inverted pendulum). The design process includes:

1. Deriving the nonlinear equations of motion via energy methods.
2. Linearizing the nonlinear model about the upright equilibrium.
3. Augmenting the state using Carleman linearization to better capture nonlinear effects.
4. Designing a state-feedback controller via LQR to stabilize both the cart and pendulum.

We explain each step in detail, describing all parameters and the purpose of the design decisions.

2 Nonlinear Model and Linearization

2.1 Equations of Motion

The dynamics of the wheeled inverted pendulum are given by:

$$a \ddot{x} + b \cos \theta \ddot{\theta} - b \sin \theta \dot{\theta}^2 = F, \quad (1)$$

$$b \cos \theta \ddot{x} + c \ddot{\theta} - m_p g l \sin \theta = 0, \quad (2)$$

where:

- x is the horizontal position of the cart,
- θ is the pendulum angle (measured from the vertical),
- F is the control force applied to the cart.

The parameters are defined as:

$$a = 2m_w + m_p + \frac{2I_w}{r^2}, \quad b = m_p l, \quad c = m_p l^2 + I_p,$$

and $d = m_p g l$.

Here,

- m_w : Mass of each wheel,
- m_p : Mass of the pendulum,
- I_w : Moment of inertia of each wheel,
- I_p : Moment of inertia of the pendulum,
- l : Distance from the axle to the pendulum's center of mass,
- r : Wheel radius,
- g : Gravitational acceleration.

We also define the determinant:

$$\Delta = a c - b^2.$$

2.2 Linearization

For small deviations around the upright equilibrium ($\theta = 0$), we approximate:

$$\sin \theta \approx \theta, \quad \cos \theta \approx 1.$$

Thus, the nonlinear equations simplify to:

$$a \ddot{x} + b \ddot{\theta} = F, \tag{3}$$

$$b \ddot{x} + c \ddot{\theta} - m_p g l \theta = 0. \tag{4}$$

By solving these simultaneously, we obtain expressions for $\ddot{x}$ and $\ddot{\theta}$. The state vector is defined as:

$$x = \begin{bmatrix} x \\ \dot{x} \\ \theta \\ \dot{\theta} \end{bmatrix}.$$

Thus, the state-space representation is:

$$\dot{x} = A_{\text{orig}} x + B_{\text{orig}} F,$$

with

$$A_{\text{orig}} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & -\frac{b m_p g l}{\Delta} & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & \frac{a m_p g l}{\Delta} & 0 \end{bmatrix}, \quad B_{\text{orig}} = \begin{bmatrix} 0 \\ \frac{c}{\Delta} \\ 0 \\ -\frac{b}{\Delta} \end{bmatrix}.$$

3 Augmentation via Carleman Linearization

To better capture the nonlinear behavior (e.g., the $\sin \theta$ nonlinearity) and enhance the controller performance, we introduce an auxiliary state z_5 . This state is intended to represent θ more accurately within the controller. The dynamics of z_5 are modified as:

$$\dot{z}_5 = \dot{\theta} + \alpha(\theta - z_5) + \delta F,$$

where:

- $\alpha > 0$ is a tuning parameter that forces z_5 to track θ ,
- $\delta > 0$ is a small constant that injects a direct influence of the control input F into the dynamics of z_5 .

The augmented state vector is then:

$$z = \begin{bmatrix} x \\ \dot{x} \\ \theta \\ \dot{\theta} \\ z_5 \end{bmatrix}.$$

The augmented state-space model becomes:

$$\dot{z} = A_{\text{aug}}z + B_{\text{aug}}F,$$

with:

$$A_{\text{aug}} = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & -\frac{b m_p g l}{\Delta} & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & \frac{a m_p g l}{\Delta} & 0 & 0 \\ 0 & 0 & \alpha & 1 & -\alpha \end{bmatrix}, \quad B_{\text{aug}} = \begin{bmatrix} 0 \\ c \\ \frac{\Delta}{\Delta} \\ 0 \\ -\frac{b}{\delta} \end{bmatrix}.$$

This augmentation via Carleman linearization "lifts" the original 4-state model into a 5-state model, enhancing the representation of the nonlinear behavior.

4 LQR Controller Design

The control objective is to design a state-feedback controller:

$$F = -Kz,$$

that minimizes the cost functional:

$$J = \int_0^\infty (z^\top Q z + F^\top R F) dt.$$

We choose the weighting matrices to place a higher penalty on deviations in the cart position:

$$Q = \text{diag}(q_1, q_2, q_3, q_4, q_5) = \text{diag}(100, 1, 100, 1, 10), \quad R = 1.$$

The optimal gain K is computed by solving the Algebraic Riccati Equation (ARE):

$$A_{\text{aug}}^\top P + P A_{\text{aug}} - P B_{\text{aug}} R^{-1} B_{\text{aug}}^\top P + Q = 0,$$

and then,

$$K = R^{-1} B_{\text{aug}}^\top P.$$

The resulting closed-loop dynamics are:

$$\dot{z} = (A_{\text{aug}} - B_{\text{aug}} K) z.$$

5 Simulation and Results

Using initial conditions with a small pendulum deviation (e.g., $\theta(0) = 0.1$ rad) and setting $z_5(0) = \theta(0)$, the closed-loop system is simulated over a finite time interval (e.g., 10 seconds). The simulation shows:

- The cart position x converges to zero.
- The pendulum angle θ is stabilized.
- The auxiliary state z_5 tracks θ appropriately.

These results confirm that increasing the weight on the cart position in Q effectively penalizes horizontal deviations, ensuring full stabilization of the system.

6 Summary of the Control Design Process

1. **Model Derivation and Linearization:** The nonlinear equations derived via energy methods are linearized about $\theta = 0$ to obtain a 4-state representation.
2. **State Augmentation:** An auxiliary state z_5 is introduced using Carleman linearization. Its dynamics:

$$\dot{z}_5 = \dot{\theta} + \alpha(\theta - z_5) + \delta F,$$

allow the augmented model to capture additional nonlinear effects.

3. **Augmented State-Space Model:** The system is lifted to a 5-state model with:

$$z = \begin{bmatrix} x \\ \dot{x} \\ \theta \\ \dot{\theta} \\ z_5 \end{bmatrix},$$

and corresponding matrices A_{aug} and B_{aug} .

4. **LQR Design:** The LQR controller is designed to minimize a cost function that heavily penalizes deviations in the cart position. The optimal gain K is computed by solving the ARE.
5. **Closed-Loop Simulation:** The control law $F = -Kz$ is applied to the augmented system. Simulation results verify that both the pendulum and the cart are stabilized.

7 Conclusion

This approach demonstrates a robust method for controlling an underactuated system:

- The initial nonlinear model is linearized around the upright equilibrium.
- Carleman linearization is used to augment the state with an auxiliary variable z_5 to capture nonlinear behavior.
- An LQR controller is designed on the augmented system, with a modified cost function that places a higher weight on the cart position.

Simulation results confirm that this hybrid method can achieve full stabilization of both the pendulum and the cart, making it a promising strategy for robust control of Segway-like systems.